

CS5211/eDP to LVDS converter design circuit diagram

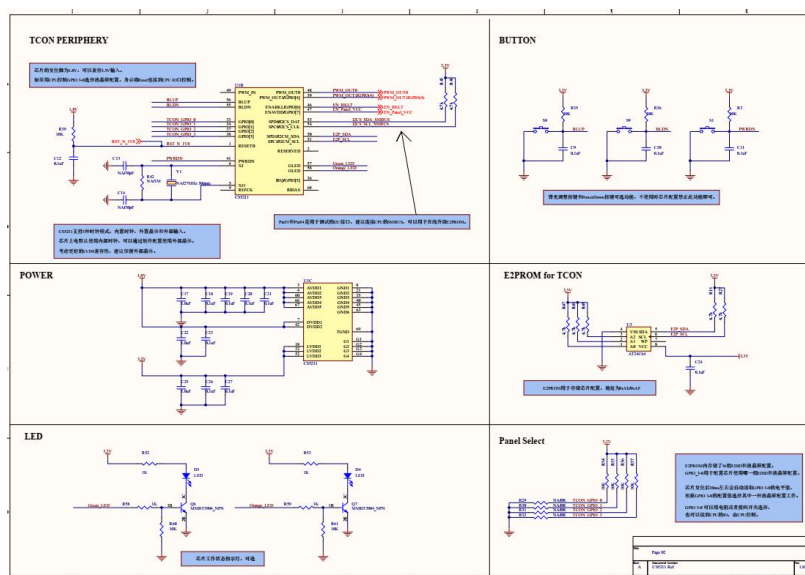
CS5211 is used to design DP to [LVDS](#) converters, DP to LVDS control boards, DP to LVDS adapter boards and other product designs. Its performance and parameters [can](#) replace and be compatible with PS8622, PS8625, CH7511 and other solution chips.

CS5211 is configured as an eDP to LVDS converter; the LVDS transmitter supports single-port 18-bit, single-port 24-bit, dual-port 18-bit and dual-port 24-bit transmission, and supports OpenLDI and SPWG bit mapping.

CS5211 is an eDP to LVDS converter with flexible configuration for low-cost display systems. CS5211 complies with eDP 1.2 and supports single and dual channel modes with speeds of 1.62Gbps and 2.7Gbps per channel. With powerful SerDes technology, CS5211 can recover high-speed serial data with low bit error rate.

The CS5211 has 4 configuration pins that can support 16 different combinations of panel resolutions and LVDS operating modes with one [EEPROM](#) image. A convenient editing tool is also provided to generate and update the EEPROM image for custom configuration.

By optimizing the design, using CS5211 can help save BoM cost. The clock source is integrated into this chip, which can save external crystals; it supports a wide range of power (core power is 1.8~1.2V), which can save power on-board equipment. The total power of CS5211 is less than 300mW, and the power supply network design can be simplified. It is easy to integrate CS5211 into popular low-cost display systems.



DP to LVDS Chip Comparison (瑞奇达代理商付S 18027661972 Q: 1540182856)						
	Function	PS8622	PS8625	CH7511	CS5211	Note
Almost Same	DP lane	1lane	2lane	2lane	2lane	
	Link Speed	2.7G max	2.7G max	2.7G max	2.7G max	
	Link Training	Fastlink+ Nolink Fulllink	Fastlink+ Nolink Fulllink	Fastlink, Fulllink	Fulllink	
	Max resolution	1680x1050@60Hz	1920x1200@60Hz	1920x1200@60Hz	1920x1200@60Hz	
	DynamicRefreshRate	Support	Support	Support	Support	
	Alternate Scrambler Seed Reset	Support	Support	Support	Support	
	LVDS interface	1 Port	2 Port	2 Port	2 Port	
	LVDS format	VESA or JEIDA	VESA or JEIDA	VESA or JEIDA	VESA or JEIDA	
	LVDS SSC	Support	Support	Support	Support	
	PWM	Passthrough and On-Chip PWM	Passthrough and On-Chip PWM	Passthrough and On-Chip PWM	Passthrough and On-Chip PWM	
CrystalFree	Yes	Yes	No	Yes		
HDCP	Support	Support	NA	NA	No use	
Firmware-less	Yes-Maybe need EDID 1kbit E2PROM	Yes-Maybe need EDID 1kbit E2PROM	No-64kbit E2PROM	No-64kbit E2PROM		
Different	Power	3.3V + 1.2V or 3.3V + integrated CDC (Need Inductor)	3.3V + 1.2V or 3.3V + integrated CDC (Need Inductor)	3.3V + 1.8V	3.3V(2.5-3.3) 1.8V(1.2-3.3V) Can support 3.3V single	
	Package	48pin 7x7	56pin 7x7	68 Pin 8x8	68 Pin 8x8	
	AUX to DDC	Yes	Yes	No	No	eePROM
	Panel Config by pin	Yes	Yes	No	No	
	16 EDID Select by pin	No	No	Yes	Yes	
	Gamma correction	NA	NA	Support	Support	No use
	Dithering 6bit input to 8bit	No	No	Yes	Yes	No use
	Back light UP/DN pin	No	No	Yes	Yes	Some client
	OSD	No	No	Yes	No	Some client

CS5211 performance description:

2-channel DisplayPort v1.1 compatible receiver

Supports 18-bit single-port, 18-bit dual-port, 24-bit single-port and 24-bit dual-port LVDS output

Supports both OpenLDI and SPWG bitmap

Built-in oscillator, no external crystal required

Embedded Linear Dropout Regulator (LDO) on-chip MCU

Select supported panels via GPIO pin control

Automatically load the boot ROM after powering on

Update serial boot ROM data via I2C bus or auxiliary channel

Automatic chip power mode control.

EMI Reduction for eDP and LVD

Wide core power range from 1.8V to 1.2V.

[QFN](#)68 package.

HBM 4KV

DP Receiver:

Compliant with embedded DisplayPort (eDP) specification 1.2

Supports 1 or 2 main link channels at 1.62Gbps or 2.7Gbps link rates.

Input in RGB format with a color depth of 18/24 bits per pixel.

Supports enhanced frame mode.

Supports VESA and CEA timing standards, resolution up to 1920x1200, 60Hz, 24-bit per pixel mode. Supports dynamic refresh rate switching.

Supports quick link training and full link training.

Supports eDP authentication: optional scrambling seed reset and optional framing.

Supports HPD interrupt.

LVDS output:

Supports 18-bit single-port, 18-bit dual-port, 24-bit single-port, and 24-bit dual-port LVDS output interfaces. Supports OpenLDI and SPWG bit mapping for LVDS applications.

When the input video is not ready, the LVDS output is kept on.

Flexible LVDS output pin swapping.

Programmable swing/common mode voltage.

Output slew rate control to reduce electromagnetic interference.

Video Processing:

Supports gamma correction.

Supports dithering and 6-bit+FRC.

Support dynamic backlight control.